Cold Gas Jet Reaction Control System
State-Space Control

Reaction control systems are used in rockets to ensure that the vehicle stays on its intended trajectory and does not spin uncontrollably. The system below uses cold gas jets to control this undesired rolling. This is achieved by using thrusters that are engaged with a solenoid actuator. There are 2 on either side of the rocket pointed in opposite directions, which allows it to control the roll in both directions.

Develop the mathematical model for the system below and design a state-space controller that meets the given design requirements.

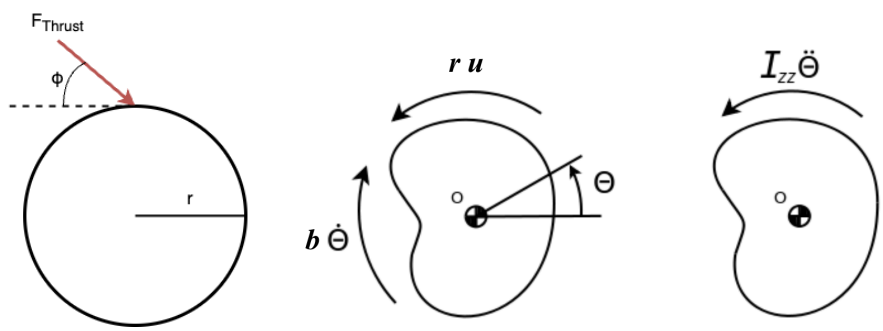


Figure 1: Free body diagram of the rocket RCS system.

Mathematical Model

First, we need to calculate the tangential component of the thrust force, starting with the diagram on the left side of Figure 1. Two thrusters are influencing the spin in either direction, so the input can be calculated as follows:

$$u = F = 2F_{thrust}cos(\phi)$$

Where F_{thrust} is the force generated by each nozzle, F is the resultant force tangential to the rocket, and ϕ is the angle between these two forces.

Next, we can sum the torques on each of the diagrams on the right side of Figure 1 and set them equal.

$$I_{zz}\ddot{\theta} = ru - b\dot{\theta}$$

Where I_{zz} is the mass moment of inertia, r is the radial distance from the rocket's center of gravity to the nozzle, b is the damping coefficient, and θ is the output. Plugging in the input and rearranging yields the following mathematical model:

$$I_{zz}\ddot{\theta} + b\dot{\theta} = 2r\,cos(\phi)\,F_{thrust} \qquad [1]$$

State-Space Representation

The state variable form can be developed using equation 1. The dependent variable is θ, and this equation is second-order, so we will have two state variables.

$$x_1 = \theta \qquad x_2 = \dot{\theta}$$

Substituting these into equation 1 yields:

$$I_{zz}\ddot{\theta} + b\,x_2 = 2r\,cos(\phi)\,F_{thrust} \qquad [2]$$

Taking the derivative of the state variables and rearranging equation 2 yields the state variable equations in standard form.

$$\dot{x_1} = x_2$$

$$\dot{x_2} = \frac{-b}{I_{zz}}\,x_2 + \frac{2r\,cos(\phi)}{I_{zz}}F_{thrust}$$

Expressed in matrix form, our [A] and [B] matrices are:

$$\begin{Bmatrix} \dot{\theta} \\ \ddot{\theta} \end{Bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & \frac{-b}{I_{zz}} \end{bmatrix} \begin{Bmatrix} \theta \\ \dot{\theta} \end{Bmatrix} + \begin{bmatrix} 0 \\ \frac{2r\,cos(\phi)}{I_{zz}} \end{bmatrix} F_{thrust}$$

The output is the angular position, θ; thus, the output matrix [D] is:

$$D = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{Bmatrix} \theta \\ \dot{\theta} \end{Bmatrix}$$

Plugging in the known values, we get the following state-space matrices:

Known Values: $I_{zz} = 0.08594\ m^2/kg$
$b = 0.002$
$r = 0.0838\ m$
$\phi = 40.75$ *degrees*
$F_{thrust} = 16.5\ N$

$$A = \begin{bmatrix} 0 & 1 \\ 0 & -0.0233 \end{bmatrix} \qquad B = \begin{bmatrix} 0 \\ 1.4778 \end{bmatrix}$$

$$D = \begin{bmatrix} 1 & 0 \end{bmatrix}$$

Calculate the Gain Matrix K Using the Design Requirements

Now, we will design a state-space controller to achieve a rise time of 0.02 seconds and no overshoot. We will determine the desired root locations using a first-order system for the first root (because we have no overshoot) and placing the second root far enough away so that it does not affect the response.

For a first-order system, $t_r = \dfrac{2.2}{a}$ where ***a*** is the location of the desired root.

We will place the second root a factor of 10 to the left of ***a***, which gives the following root locations:

$$a = \frac{2.2}{t_r} \quad \Rightarrow a = 110$$

$$\alpha_c = (s + 110)(s + 1100) = 0 \qquad \text{[3a]}$$

$$s^2 + 1210\, s + 121000 = 0 \qquad \text{[3b]}$$

Now, we get the characteristic equation from our matrices.

$$|s[I] - [A] + [B][K]| = 0 \qquad \text{[4a]}$$

$$|s \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ 0 & -0.0233 \end{bmatrix} + \begin{bmatrix} 0 \\ 1.4778 \end{bmatrix} [K_1 \;\; K_2]| = 0$$

$$| \begin{bmatrix} s & 0 \\ 0 & s \end{bmatrix} - \begin{bmatrix} 0 & 1 \\ 0 & -0.0233 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 1.4778\, K_1 & 1.4778\, K_2 \end{bmatrix} | = 0$$

$$\begin{vmatrix} s & -1 \\ 1.4778\, K_1 & (s + 0.0233 + 1.4778\, K_2) \end{vmatrix} = 0$$

$$s^2 + (0.0233 + 1.4778\, K_2)s + 1.4778\, K_1 = 0 \qquad \text{[4b]}$$

Equating terms from equations [3b] & [4b] gives the following K matrix:

$$1.4778\, K_1 = 121000$$
$$0.0233 + 1.4778\, K_2 = 1210$$

$$K = \begin{bmatrix} 81880 & 818.8 \end{bmatrix}$$

Closed Loop Matrix [A] Step Response

The closed loop [A] matrix can now be calculated, and the system response can be plotted with this new closed loop [A].

$$A_{cl} = [A] - [B]\,[K]$$

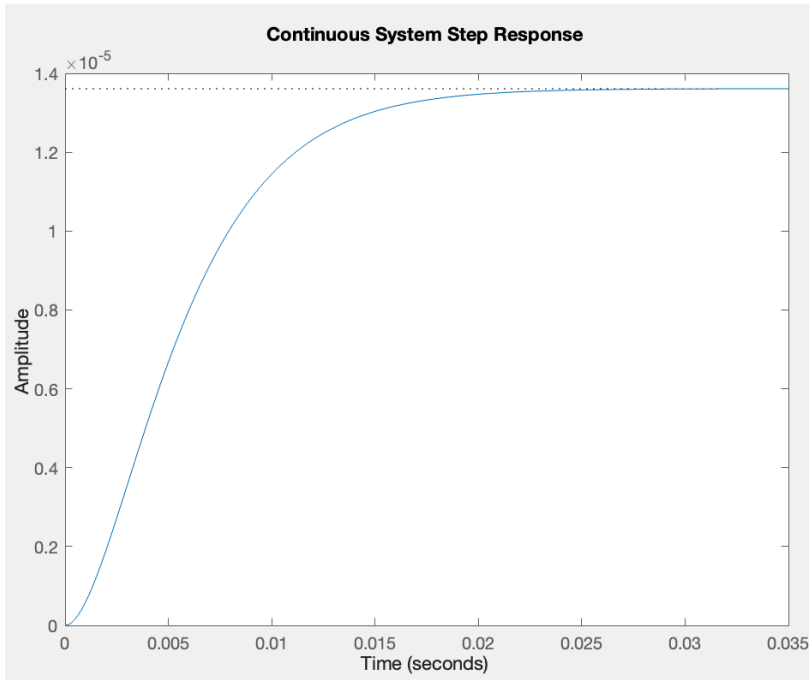
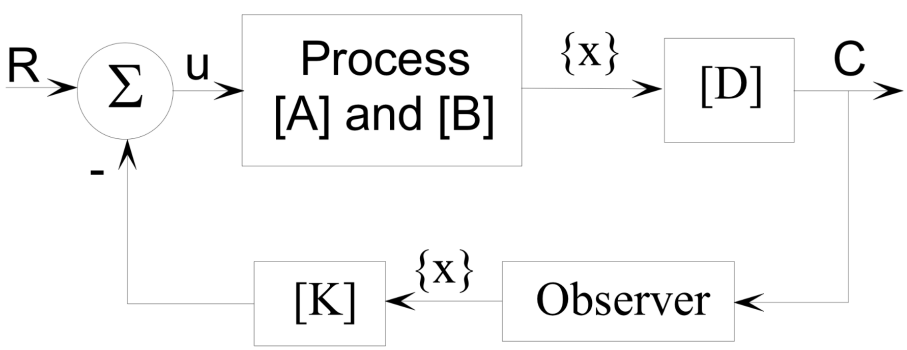


Figure 2: The closed loop step response with state-space control.

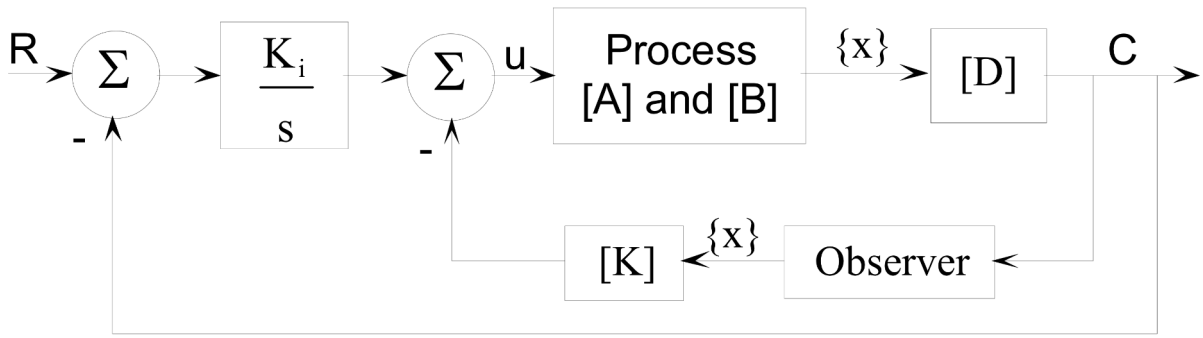
This step response shows that the controller meets the rise time requirements, but the amplitude should go to unity. This tells us that there is a large amount of steady state error. This can be dealt with by adding an integrator and outer feedback loop.

Adding An Integrator and Outer Feedback Loop

The block diagram for the system with state feedback:



We need to add an integrator in series with the state feedback, along with an outer feedback loop. The new block diagram is:



The addition of the integrator creates another state variable because it increases the type of the system by one. This new state variable is:

$$\dot{x_N} = R - [D]\,\{x\}$$

This gives us new state variable equations that are as follows:

$$\{\dot{x}\} = [A]\{x\} + [B]\,u$$

$$\dot{x_N} = R - [D]\{x\}$$

$$C = [D]\{x\}$$

In matrix form, these equations give us the new state-space matrices [Ai], [Bi], and [Di]:

$$[A_i] = \begin{bmatrix} [A] & \{0\} \\ -[D] & 0 \end{bmatrix} \qquad [B_i] = \begin{Bmatrix} [B] \\ 0 \end{Bmatrix} \qquad [D_i] = \begin{bmatrix} [D] & 0 \end{bmatrix}$$

Plugging in our original state-space matrices:

$$[A_i] = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -0.0233 & 0 \\ -1 & 0 & 0 \end{bmatrix} \qquad [B_i] = \begin{Bmatrix} 0 \\ 1.4778 \\ 0 \end{Bmatrix} \qquad [D_i] = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$$

To find the new gain matrix, Ki, we repeat the processes that lead to equations [3b] and [4b]. However, now that we have another state variable and a third-order system, we need to include another root. Since the second root was already placed far away, we can place the third one right next to it.

$$\alpha_i = (s + 110)(s + 1100)(s + 1101) = 0 \qquad [5]$$

Using equation [4a] with the new SS matrices gives us something to equate this to, which gives us the new gain matrix, Ki. This part was done in Matlab using the place command.

```
%New gain matrix Ki
pi = [-110 -1100 -1101];
Ki = place(Ai,Bi,pi);
Ki(end) = -Ki(end)
```

$$K_i = \begin{bmatrix} 983376 & 1564 & 90.15e6 \end{bmatrix}$$

The closed loop state feedback matrices are then calculated using the following equations:

$$A_i cl = \begin{bmatrix} [A] - [B][K_{i1}, K_{i2}] & [B]K_{i3} \\ -[D] & 0 \end{bmatrix} \qquad B_i cl = \begin{Bmatrix} \{0\} \\ 1 \end{Bmatrix}$$

Where Ki_1, Ki_2, and Ki_3 are the first, second, and third terms of Ki, respectively.
The closed loop matrix [Dcl] is the same as [Di]. Thus, our final closed loop matrices are as follows:

$$A_i cl = \begin{bmatrix} 0 & 1 & 0 \\ -1.453e6 & -2311 & 1.332e8 \\ -1 & 0 & 0 \end{bmatrix} \qquad B_i cl = \begin{Bmatrix} 0 \\ 0 \\ 1 \end{Bmatrix}$$

$$D_i cl = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$$

Closed Loop Response with Added Integrator and Outer Feedback Loop

The response to a step input of this closed loop state space controller is as follows:

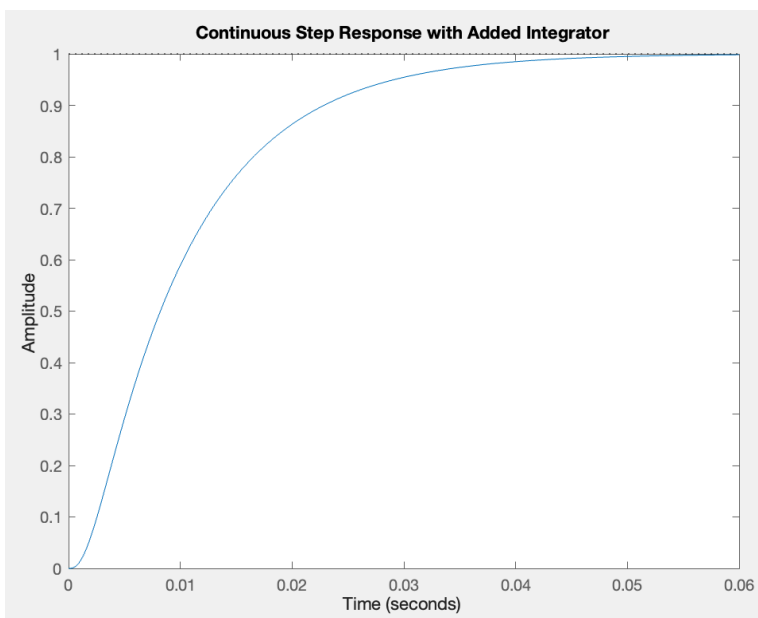


Figure 3: The step response with an added integrator.

From the step response in Figure 3, we can see that the system has a rise time of 0.0203 seconds and no overshoot. This state-space controller now satisfies the design requirements.

Matlab Code

```
clear,clc
close all
format compact

%Constants
% The actual values will differ from these, but I will have to determine
% the actual values experimentally once the rocket is assembled.

Izz = 0.08594; %Moment of inertia about z axis
b = 0.002; %The damping coefficient
r = 3.3*2.54/100; %radial distance from z axis to thruster converted to meters
phi = 40.749; %angle of thrusters from tangent
F_thrust = 16.5; %The thrust from the nozzles

%The state space matrices
A = [0 1; 0 -b/Izz];
B = [0; 2*cosd(phi)*r/Izz];
C = [1 0];
D = 0;

%state space system
sys = ss(A,B,C,D)
figure
pzmap(sys)

%Design requirements
tr = 0.02;
a = 2.2/tr;
p = [-a -a*10];
K = place(A,B,p)

%Closed loop A matrix
Acl = A-B*K;
eig(Acl);

%Closed loop step response
sys_cl = ss(Acl,B,C,D)
figure
step(sys_cl)
title('Continuous System Step Response')

%Steady state error is not zero, so need to add integrator and outer
%feedback loop
Ai = [A [0;0];-C 0];
Bi = [B;0];
Ci = [C 0];
Di = 0;
```

```
%New gain matrix Ki
pi = [p p(2)-1];
Ki = place(Ai,Bi,pi);
Ki(end) = -Ki(end)

%New Closed Loop matrices
Ai_cl = [A-B*Ki(1:(length(Ki)-1)) B*(Ki(length(Ki)));-C 0];
Bi_cl = [0;0;1];
Ci_cl = Ci;
Di_cl = 0;

%New closed loop system
sys_i = ss(Ai_cl,Bi_cl,Ci_cl,0)
figure
step(sys_i)
title('Continuous Step Response with Added Integrator')
```

Digital State-Space Controller in Matlab
Code:

```
clear,clc
close all

%Constants
% The actual values will differ from these, but I will have to determine
% the actual values experimentally once the the rocket is assembled.

Izz = 0.08594; %Moment of inertia about z axis
b = 0.002; %The damping coefficient
r = 3.3*2.54/100; %radial distance from z axis to thruster converted to meters
phi = 40.749; %angle of thrusters from tangent
F_thrust = 16.5; %The thrust from the nozzles

%The state space matrices
A = [0 1; 0 -b/Izz];
B = [0; 2*cosd(phi)*r/Izz];
C = [1 0];
D = 0;
sys = ss(A,B,C,D)
figure
pzmap(sys)

%Convert to digital
Ts = 0.001;
sysd = c2d(sys,Ts,'ZOH');
figure
pzmap(sysd)
```

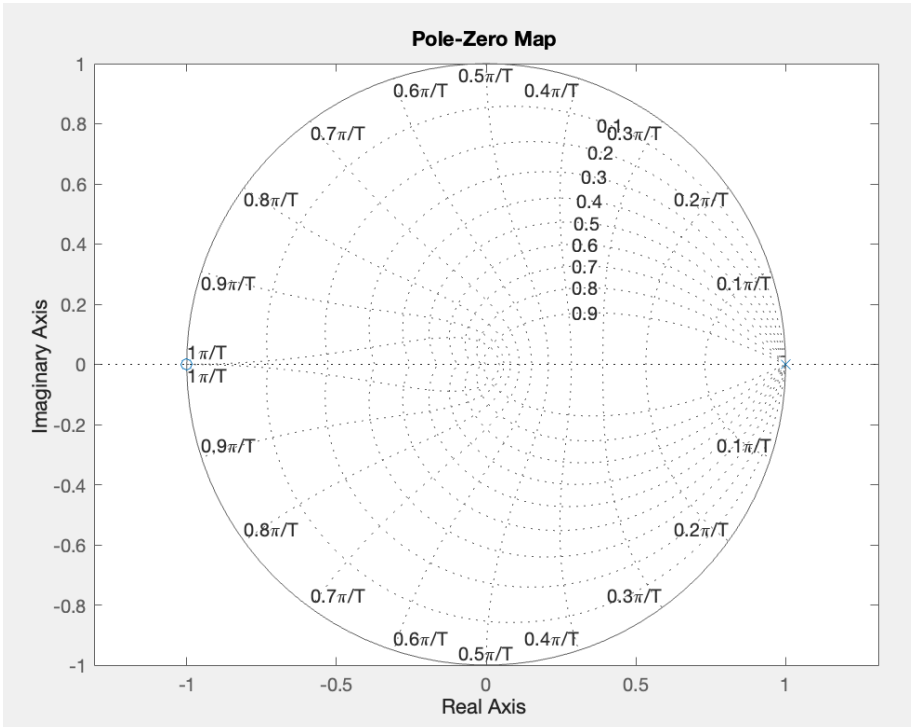
```
[poles_d zeros_d] = pzmap(sysd);
zgrid
axis equal

%Design requirements
tr = 0.02;
PMO = 0.5;
zeta = zeta_PMO(PMO);
wn = wn_riseTime(tr,zeta);
wd = sqrt(1-zeta^2)*wn;
ws = 2*pi/Ts;
[Z_mag,Z_ang] = Z_magAngle(zeta,wd,ws); %uses custom function
z1 = Z_mag*cos(Z_ang)+Z_mag*sin(Z_ang)*i;
z2 = Z_mag*cos(Z_ang)-Z_mag*sin(Z_ang)*i;

%First CL system
p = [z1; z2];
K1 = place(sysd.A,sysd.B,p);
Acl1 = sysd.A-sysd.B*K1;
SYS1 = ss(Acl1,sysd.B,sysd.C,sysd.D,Ts);
figure
step(SYS1)
axis auto
title('Initial Closed Loop System Step Response')
figure
pzmap(SYS1)
axis equal

%Add integrator and outer feedback loop
pd = [z1;z2;0.2];
A = sysd.A;
B = sysd.B;
C = sysd.C;
Ahat = [A B;zeros(length(A),1)' 0];
Bhat = [zeros(length(A),1);1];
Khat = place(Ahat,Bhat,pd)
AA = [A-eye(size(A)) B; C*A C*B]
KK = [Khat + [zeros(length(A),1)' 1]]*inv(AA);
K2 = KK(1:end-1);
K1 = KK(end);
ACL = [A-B*K2 B*K1; -C*A+C*B*K2 1-C*B*K1];
BCL = Bhat;
CCL = [C 0];
sys_CLi = ss(ACL,BCL,CCL,0,Ts)
figure
step(sys_CLi,0.03)
ylim([0 1.2])
title('Step Response with Integrator Feedback')
```

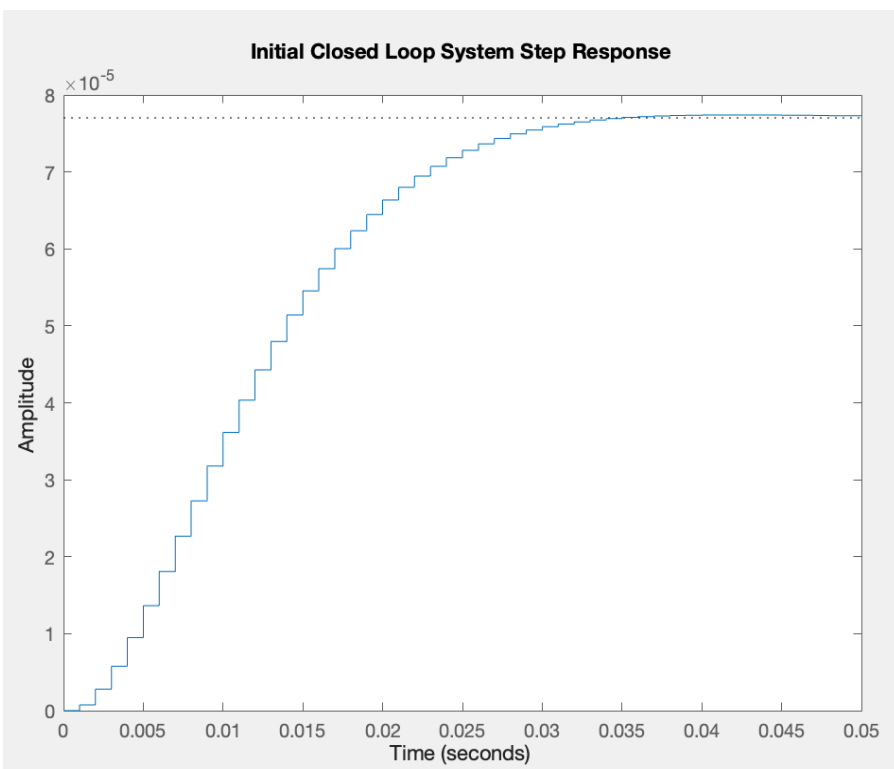
Output to Command Window

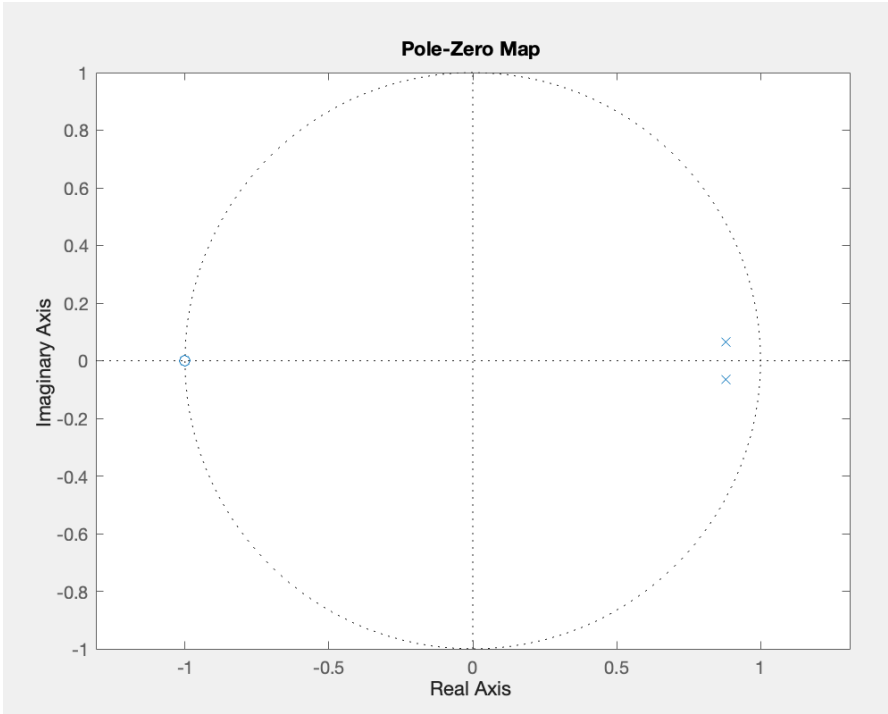
```
sys =

  A =
              x1        x2
   x1          0         1
   x2          0  -0.02327

  B =
          u1
   x1      0
   x2  1.478

  C =
       x1  x2
   y1   1   0

  D =
       u1
   y1   0

Continuous-time state-space model.


sysd =

  A =
          x1     x2
   x1      1  0.001
   x2      0      1

  B =
              u1
   x1  7.389e-07
   x2   0.001478

  C =
       x1  x2
   y1   1   0

  D =
       u1
   y1   0

Sample time: 0.001 seconds
Discrete-time state-space model.
```

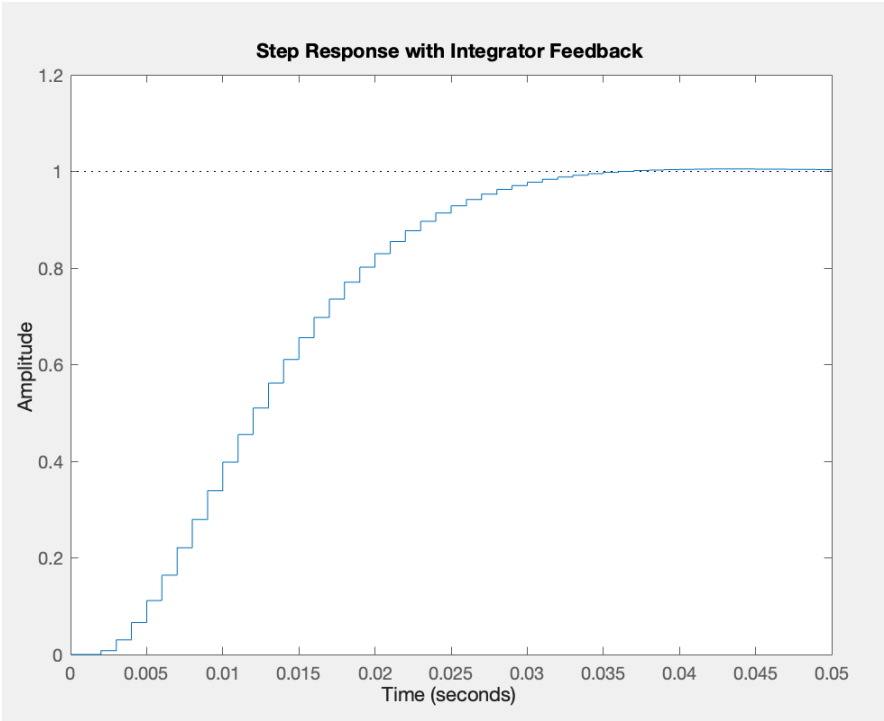
```
SYS1 =

  A =
              x1          x2
   x1      0.9904  0.0008831
   x2      -19.19     0.7663

  B =
              u1
   x1  7.389e-07
   x2   0.001478

  C =
       x1  x2
   y1   1   0

  D =
       u1
   y1   0

Sample time: 0.001 seconds
Discrete-time state-space model.


sys_CLi =

  A =
               x1           x2           x3
   x1       0.9046   0.0005299     0.007674
   x2       -190.8     0.05976        15.35
   x3      -0.9046  -0.0005299       0.9923

  B =
       u1
   x1   0
   x2   0
   x3   1

  C =
       x1  x2  x3
   y1   1   0   0

  D =
       u1
   y1   0

Sample time: 0.001 seconds
Discrete-time state-space model.
```

Plots


Pole-Zero Map
Imaginary Axis
Real Axis
0.5π/T
0.6π/T
0.4π/T
0.7π/T
0.3π/T
0.8π/T
0.2π/T
0.9π/T
0.1π/T
1π/T
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9


Initial Closed Loop System Step Response
×10^-5
Amplitude
Time (seconds)


Pole-Zero Map
Imaginary Axis
Real Axis
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
-1
-0.5
0
0.5
1


Step Response with Integrator Feedback
Amplitude
Time (seconds)
1.2
1
0.8
0.6
0.4
0.2
0
0
0.005
0.01
0.015
0.02
0.025
0.03
0.035
0.04
0.045
0.05